

Politecnico di Torino

Department of Electronics and Telecommunications

Master's degree in Mechatronic Engineering



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Sensors, Embedded Systems and Algorithms for Service Robotics

Report 3

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1 OBJECTIVE

The aim of this laboratory activity is to ...

- Develop a ROS2 package to run control algorithms on a mobile robot.
- Obstacle avoidance and dynamic goal chasing with DWA.
- Test the developed algorithms on the real robots following an AprilTag detected through RGB camera.

2 DESIGN

2.1 Task 1

The primary objective of Task 1 is to design and implement a ROS 2 navigation node for a mobile robot capable of dynamic target following and obstacle avoidance using the Dynamic Window Approach (DWA). The navigation algorithm is implemented within a ROS 2 node named *Follow_Target*, which runs at a fixed frequency of 15 Hz through a timer-based callback that emulates a real-time control loop.

The node subscribes to the */scan* topic to acquire LiDAR measurements, to */dynamic_goal_pose* for the target position, and to */ground_truth* for the robot state. The resulting optimal linear and angular velocity commands are published to the */cmd_vel* topic.

At each control iteration, admissible velocity pairs (v, ω) are sampled within the robot's dynamic window. For each candidate pair, the robot's future pose is predicted over a time horizon of $T=0.3s$. When the angular velocity is negligible ($|\omega| < 10^{-5}$), a linear motion model is used; otherwise, a circular arc model is applied to compute the predicted position (x_{new}, y_{new}) .

Trajectory selection is driven by an objective function J , which evaluates each candidate velocity pair by combining three terms: heading alignment, linear velocity, and obstacle distance. The cost function is defined as

$$J = \alpha \cdot \text{heading} + \beta \cdot v + \gamma \cdot \frac{1}{\text{dist}_{\text{obst}}}$$

where the weights were empirically tuned to $\alpha=0.8$, $\beta=-0.5$, and $\gamma=0.5$. The heading term measures the normalized angular difference between the target direction and the predicted motion direction, the velocity term favors higher forward speeds, and the obstacle distance term penalizes trajectories that approach obstacles too closely. At each iteration, the velocity pair yielding the minimum cost is selected and sent to the robot.

LiDAR data preprocessing is performed to ensure robust obstacle perception. Raw scan measurements are filtered to handle sensor noise and range limits. NaN values are replaced with a minimum safety distance of 0.15 m, while infinite values are clamped to the maximum measurable range of 3.5 m. To reduce computational complexity, the full 360° scan is downsampled into 20 angular sectors, and for each sector only the minimum distance is retained. These distances are converted into Cartesian coordinates in the robot frame and used during trajectory evaluation.

Safety is enforced through a dual-layer mechanism. During cost evaluation, any candidate velocity leading to a predicted obstacle distance below 0.15 m is immediately discarded, and an additional braking-distance constraint ensures that the robot can safely decelerate given its maximum acceleration. In parallel, an emergency stop mechanism monitors the raw LiDAR data, and if any measurement falls below a threshold of 0.2 m, the robot is immediately halted and a warning is issued.

2.2 Task 2

For Task 2, the core structure of the Dynamic Window Approach was preserved, including velocity sampling, trajectory prediction, cost evaluation, and command selection. The main design modifications were introduced in the objective function to improve behavior during goal approach and dynamic target following.

For each candidate velocity pair (v, ω) , the robot's future pose (x_{new}, y_{new}) is predicted, and the corresponding distance to the target is computed as

$$d_{pred} = \sqrt{(x_{goal} - x_{new})^2 + (y_{goal} - y_{new})^2}$$

This predicted distance is used to activate two additional behaviors. To enable a smooth deceleration near the target, a distance threshold $dist_{threshold}$ is defined. When d_{pred} falls below this threshold, an additional velocity-related cost term vel' is introduced. This term is designed as a function of the linear velocity and the proximity to the target, such that its contribution through the unchanged velocity weight $\beta = -0.5$ penalizes aggressive commands when the robot is close to the goal, favoring lower speeds during the final approach.

To support dynamic target following, an additional cost term is introduced to regulate the distance between the robot and the target. This term penalizes deviations from a desired following distance d_{des}

$$Dist_{target} = (d_{pred} - d_{des})^2$$

Weighted by a coefficient δ , this term is added to the overall cost function, allowing the DWA to optimize not only heading alignment and obstacle avoidance but also the stability of the robot–target distance during motion.

The weighting parameters were selected through experimental tuning. The heading and obstacle avoidance weights α and γ were kept equal to those used in Task 1, as they provided stable behavior when combined with the new terms. The parameter δ proved to be the most influential, directly controlling the priority given to distance regulation. Higher values of δ encouraged trajectories that maintained the desired separation from the target, reduced freezing behavior in narrow or difficult areas, and prevented overly conservative obstacle-avoidance strategies that could cause loss of target visibility. Based on these observations, a relatively high value of $\delta = 1.5$.

2.3 Task 3

For Task 3, the core Dynamic Window Approach developed in simulation was preserved, while several design adaptations were introduced to enable reliable execution on the real robot. The most significant change concerned target acquisition. Unlike simulation, where the target position is available through ground-truth information, the real robot must reconstruct the goal from camera data. Target detection is performed using the `/camera/Landmarks` topic, from which the range and bearing of the detected AprilTag are extracted and converted into Cartesian coordinates in the robot frame $(X_{goal_robot}, Y_{goal_robot})$. Using the robot pose obtained from `/odom`, this estimate is then transformed into the odometry frame, producing a consistent goal representation over time. During each control iteration, the relative goal position is recomputed using the stored goal pose and the current robot pose, ensuring that the geometry used by the cost function remains consistent with the formulation adopted in Task 2.

Sensor handling was also adapted to account for real-world LiDAR behavior. In contrast to simulation, laser scans on the real robot frequently contain NaN and infinite values. To prevent these readings from being misinterpreted as obstacles and blocking the planner, an explicit scan-cleaning step was introduced. Invalid measurements are replaced with large distance values before further processing. After cleaning, the LiDAR scan is downsampled into a fixed number of angular sectors, and for each sector the minimum range is selected and converted into Cartesian coordinates in the robot frame. This compact obstacle representation allows efficient evaluation of candidate trajectories without processing the full scan at every iteration.

Safety constraints were revised to suit the confined laboratory environment. The minimum distance threshold used to discard unsafe trajectories in the cost function was reduced from approximately 0.20 m to 0.10 m. The original threshold proved overly conservative, often preventing the planner from finding feasible trajectories in narrow passages and causing the robot to freeze. The reduced threshold allows navigation in tighter spaces while still maintaining an adequate safety margin. In addition, the robot's kinematic limits were scaled down, reducing maximum velocities and accelerations to better match the physical dynamics of the platform and ensure stable motion execution.

Further parameter tuning was performed to balance obstacle avoidance and target tracking in real-world conditions. The obstacle cost weight γ was significantly reduced to 0.1, as higher values led to excessive rejection of safe trajectories in narrow environments. The target-following weight δ was kept high at 1.3, slightly lower than in Task 2, to ensure that maintaining visibility and proximity to the target remained a priority. The heading and velocity weights $\alpha=0.8$ and $\beta=-0.5$ were retained unchanged, as they provided satisfactory trajectory selection without requiring further tuning.

3 EXPERIMENT

The experimental evaluation was designed to validate the proposed Dynamic Window Approach–based navigation system under progressively more realistic conditions. The experiments were organized according to the three tasks of the laboratory assignment and were conducted both in simulation and on the real robot. This structure allowed systematic verification of the algorithm’s behavior, parameter tuning, and robustness against sensor noise and environmental constraints.

3.1 SIMULATION

3.1.1 Task 1 – Obstacle Avoidance and Dynamic Target Following

The experimental evaluation of Task 1 was conducted in a simulation environment to validate the correctness and effectiveness of the proposed DWA-based navigation system. The robot was tasked with following a dynamically moving target while avoiding obstacles using the implemented controller. During the experiments, the robot pose, target position, LiDAR measurements, and velocity commands were recorded for analysis.

The experiments focused on assessing the robot’s ability to track the target, maintain safe distances from obstacles, and generate stable velocity commands under dynamic conditions. Particular attention was given to verifying the impact of the selected cost-function weights and safety constraints on trajectory smoothness, obstacle clearance, and overall navigation reliability.

3.1.2 Task 2 – Extended Cost Function for Smooth Tracking

The experimental evaluation of Task 2 focused on assessing the impact of the extended cost function on tracking stability and motion smoothness. The robot was tested in simulation using dynamic target trajectories with varying curvature and speed to highlight differences with respect to the baseline implementation.

The experiments showed that the additional velocity penalty enabled a smoother and more controlled approach to the target, reducing oscillations and abrupt stops near the goal. The target-distance regulation term significantly improved tracking stability, allowing the robot to maintain a more consistent separation from the moving target even in constrained environments and during rapid target motion. Parameter tuning confirmed the dominant role of the distance-regulation weight δ in achieving robust and reliable dynamic following.

3.2 REAL ROBOT

3.2.1 Task 3 – Real World Deployment and Validation

To validate the adapted navigation system, multiple experiments were conducted in the laboratory environment using the real robot. The target was represented by an AprilTag manually moved in front of the robot at varying speeds and along different paths, requiring the robot to track the target while navigating around obstacles in a confined space.

These experiments were designed to evaluate the robustness of the modified perception pipeline, the effectiveness of the adjusted safety constraints, and the overall stability of the DWA controller under real-world conditions. The robot's behavior was monitored during target following, obstacle avoidance, and transitions through narrow passages. The selected parameter configuration was verified to provide reliable tracking, prevent excessive freezing behavior, and maintain safe distances from obstacles throughout the experimental runs.

4 RESULT & DISCUSSION

4.1 Task 1

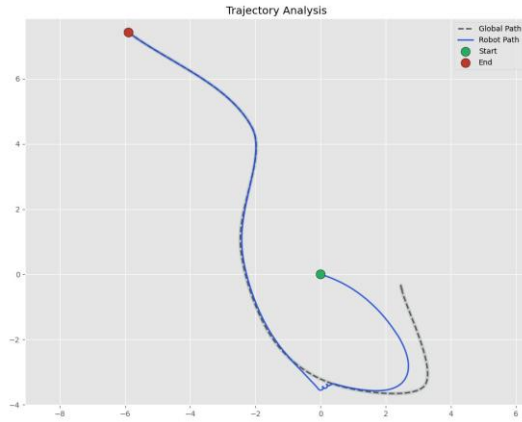


Figure 1: Trajectory Plot Of Simulated Robot

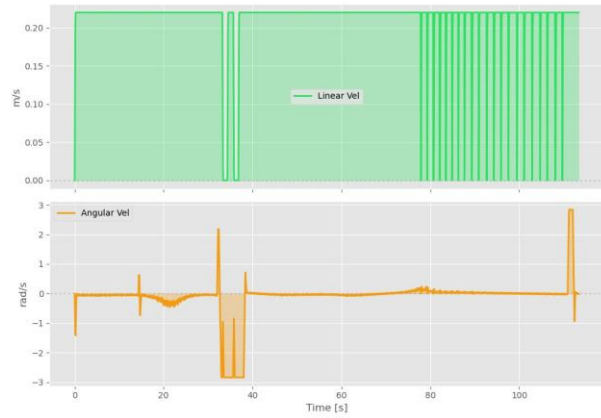


Figure 2: Velocity Plot of Simulated Robot

The trajectory plot shows that the robot accurately follows the dynamic target, starting from the origin and smoothly converging to the final goal position around $(-6, 7.5)$. The robot successfully handles wide S-shaped maneuvers, demonstrating precise path tracking and stable motion throughout the task.

The velocity profiles confirm the expected behavior of the DWA controller. The linear velocity rapidly reaches its maximum value and remains high for most of the run, with brief reductions during sharp turns where heading alignment is prioritized. Correspondingly, the angular velocity exhibits pronounced peaks during these maneuvers, indicating correct selection of aggressive rotational commands to minimize heading error. Temporary fluctuations in linear speed further reflect the controller's ability to balance

target tracking with obstacle avoidance in more constrained sections. Overall, the absence of collisions and successful convergence within the goal tolerance confirm the effectiveness of the chosen cost-function parameters.

Metric (Simulation)	Value
RMSE Bearing	0.3624 rad
RMSE Path Dist	0.9612 m
Total Duration	118.55 s
Time of Tracking % ($d < 2.0\text{m}$)	100%
Success Samples (frames)	978
Timeout Events (times > 10.0s)	4
Time of Tracking % :	100%
Collision Events (< 0.2 m)	0
Min Obstacle Dist:	0.3405 m
Avg Obstacle Dist:	2.6113 m

Table 1: Metrics of the Simulated Robot

The simulation metrics confirm the effectiveness of the proposed navigation system. The RMSE values for bearing (0.3642 rad) and path distance (0.9612 m) demonstrate accurate tracking of the target trajectory. The system achieved a Time of Tracking of 100%, indicating that the robot maintained continuous visual contact with the target for the entire duration of the 118.55-second experiment.

A total of 978 Success Samples were recorded, showing that the robot consistently operated within an acceptable distance from the dynamic goal. A small number of Timeout events (four occurrences) were observed, which can be attributed to the sharp S-shaped maneuvers in the trajectory, during which the robot temporarily reduced its forward speed to prioritize heading correction.

Most importantly, no collisions occurred during the experiment. The minimum distance to obstacles was measured at 0.3405 m, well above both the emergency stop threshold of 0.2 m and the safety margin of 0.15 m used during trajectory evaluation. These results confirm that the obstacle avoidance term and the LiDAR preprocessing strategy enabled safe and reliable navigation throughout the environment.

4.2 Task 2

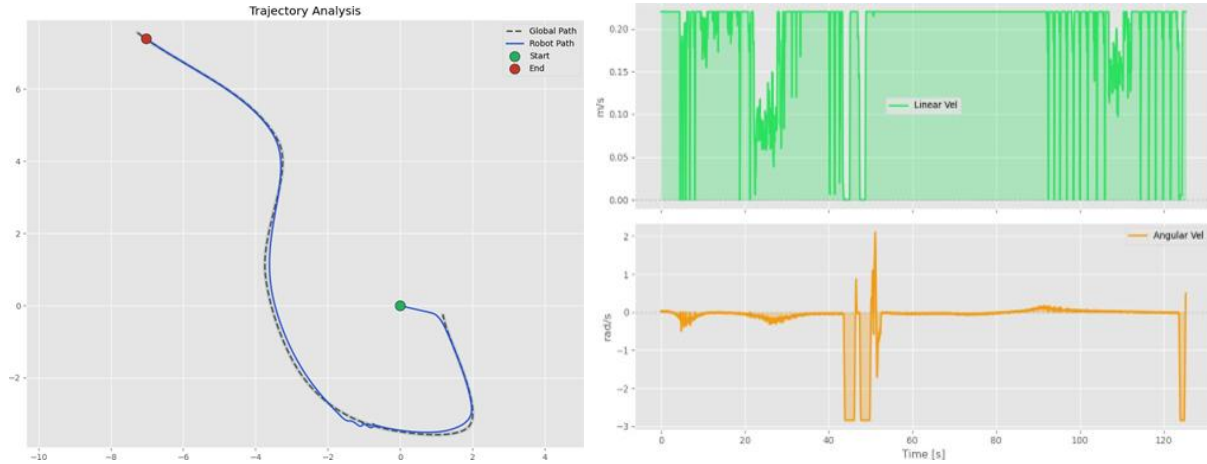


Figure 3: Trajectory & Velocity Profile Plot of the Robot operated in Simulation

The results obtained for Task 2 highlight the impact of the extended DWA cost function on both trajectory tracking and velocity behavior. From the trajectory analysis plot, it can be observed that the robot closely follows the global path of the moving target, maintaining a smooth and continuous trajectory despite the pronounced S-shaped curvature. Compared to the baseline implementation, the robot exhibits more regular spacing from the reference path, indicating improved stability in dynamic following. The start and end points confirm that the robot successfully converges toward the target without abrupt deviations, even in sections requiring significant curvature changes.

The velocity profiles further illustrate the effect of the modified cost function. The linear velocity remains generally high during straight and gently curved segments, demonstrating efficient target chasing, while noticeable reductions occur in tighter portions of the trajectory. These slowdowns reflect the activation of the distance-dependent velocity penalty, which encourages more cautious motion when the robot approaches the target or navigates constrained areas. The angular velocity profile shows pronounced peaks during sharp turns, indicating that the controller correctly prioritizes heading alignment when required. Outside these maneuvers, angular velocity remains relatively smooth and bounded, suggesting stable control behavior. Overall, the plots confirm that the additional cost terms introduced in Task 2 successfully balance tracking accuracy, smooth motion, and safety. The robot maintains a stable distance from the target while avoiding excessive oscillations in speed, demonstrating the effectiveness of the extended DWA formulation for dynamic target following.

Metric (Simulation)	Value
RMSE Bearing	0.3624 rad
RMSE Range	0.5080 m
Total Duration	115.49 s
Time of Tracking % ($d < 2.0\text{m}$)	100%
Success Samples ($d < 0.25\text{ m}$)	11
Success Samples ($d < 0.40\text{ m}$)	1327
Timeout Events ($> 15.0\text{ s}$)	5
Collision Events ($< 0.2\text{ m}$)	0

Table 2: Metrics For Task 2

The obtained metrics indicate satisfactory tracking performance, with acceptable RMSE values for both bearing and range. The Time of Tracking achieved the maximum value of 100%, meaning that the robot remained within a distance of 2.0 m from the target for the entire duration of the experiment.

The number of Success Samples computed using the strict distance threshold of $d < 0.25\text{ m}$ is relatively low, with only 11 occurrences. However, when the threshold is slightly relaxed to $d < 0.40\text{ m}$, the number of success activations increases significantly to 1327. This behavior suggests that the robot consistently followed the target at close range but remained just outside the most restrictive tolerance, which can be attributed to the lower speed of the target relative to the robot's dynamics.

A total of five Timeout events were observed during the experiment. Importantly, no collision events occurred, and the minimum distance to obstacles was measured at 0.2796 m, confirming that safe navigation was maintained throughout the trajectory.

4.3 Task 3

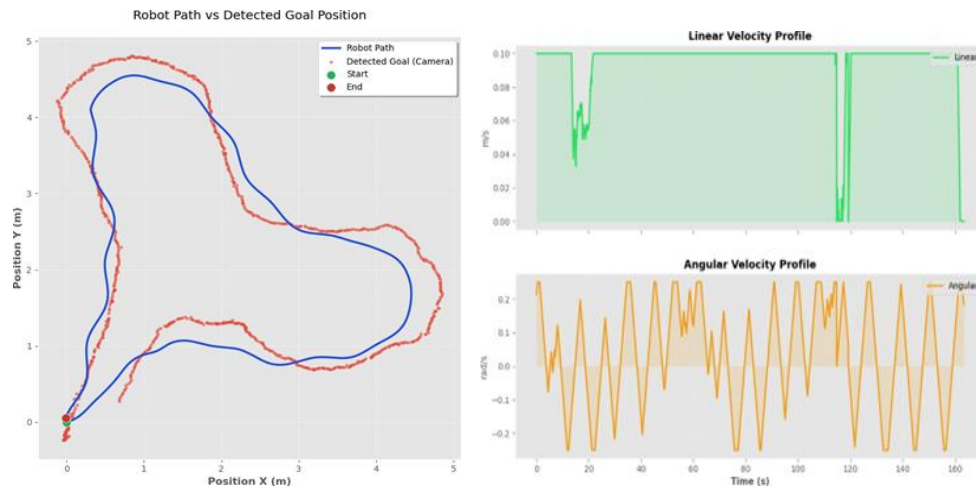


Figure 4: Trajectory & Velocity Profile of 1st Run

Metric (Simulation)	Value
RMSE Bearing	0.3624 rad
RMSE Range	0.5080 m
Total Duration	115.49 s
Time of Tracking % ($d < 2.0\text{m}$)	100%
Success Samples ($d < 0.25\text{ m}$)	11
Success Samples ($d < 0.40\text{ m}$)	1327
Timeout Events ($> 15.0\text{ s}$)	5
Collision Events ($< 0.2\text{ m}$)	0

Table 3: Data from 1st Successful Run

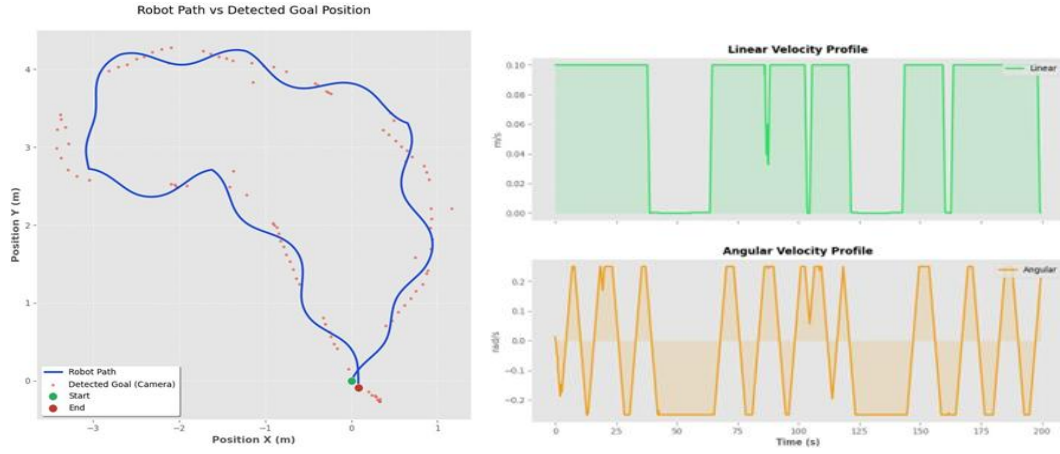


Figure 5: Trajectory & Velocity Profile of 2nd Run

Metric (Run 2)	Value
Time of Tracking % ($d < 3.0\text{m}$)	100.00 %
RMSE Bearing	0.2040 rad
RMSE Range	0.5414 m
Total Duration	198.45 s
Success Activations (15Hz)	1474
Timeout Events ($> 15.0\text{s}$)	1
Collision Events ($< 0.20\text{m}$)	0
Min / Avg Obstacle Dist	0.2120 / 3.5626 m

Table 4: Data from 2nd Successful Run

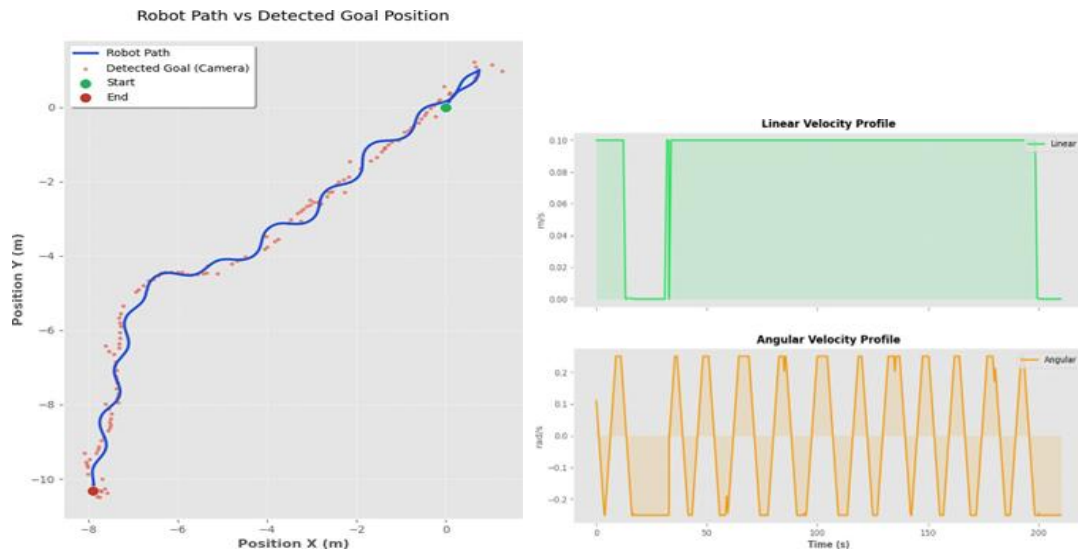


Figure 6: Trajectory & Velocity Profile of 3rd Run

Metric (Run 3)	Value
Time of Tracking % ($d < 3.0\text{m}$)	100.00 %
RMSE Bearing	0.2482 rad
RMSE Range	0.7908 m
Total Duration	215.65 s
Success Activations (15Hz)	1264
Timeout Events ($> 15.0\text{s}$)	2
Collision Events ($< 0.20\text{m}$)	30
Min / Avg Obstacle Dist	0.1530 / 3.8000 m

Table 5: Data from 3rd Successful Run

The real-robot experiments demonstrate consistent tracking performance, with all three runs achieving a 100% tracking rate within the 3.0 m threshold. Performance differences across runs were mainly influenced by environmental complexity and target trajectory characteristics.

Trajectory 1, conducted inside the laboratory classroom, represented the most challenging scenario. The complex target motion required the robot to maintain a larger separation distance to ensure safe navigation, resulting in a lower number of success activations (172). This behavior reflects a deliberate prioritization of stability and safety over close-proximity convergence in a cluttered environment.

Trajectory 2 achieved the best overall performance, with 1474 success activations and no collisions. The target remained consistently in front of the robot, minimizing bearing error and enabling stable tracking. Although a reduced rate of camera detections was observed, likely due to thermal throttling, the controller maintained reliable performance, highlighting its robustness to partial sensor degradation.

Trajectory 3 showed intermediate results, with 1264 success activations and 30 minor collision events below 0.20 m. These events occurred mainly during aggressive maneuvers when the target was not centered, requiring the DWA to dynamically balance heading alignment and obstacle avoidance.

Overall, the results confirm the robustness of the implemented DWA controller, which successfully avoided critical collisions below 0.15 m and maintained effective tracking even in narrow environments and under degraded sensing conditions.

To enable quantitative evaluation, a dedicated data logging mechanism was integrated into the control architecture. Cumulative counters for success samples, collision events, timeout events, and total control steps were published at 15 Hz on the */metrics* topic and recorded via rosbag. Offline processing using a custom Python script was then used to compute the final performance metrics, including success rate and tracking time.